Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Project: LingoLive — Understand. Reply. Keep going.

Team name: CreativeEngine

Team leader: Ashish T. Vasant | Member: Janhvi Singh

Problem: Help migrants participate in everyday conversations in an unfamiliar language.

Theme: Sustainability & Social Impact | Category: Accessibility

Brief about the idea

A conversation assistant for life in a new place.

LingoLive listens to another person, explains their words and offers three useful replies in the local language, with pronunciation. Plans, recent turns and location-linked memory make those replies personal.

Understand now

Live speech → familiar-language meaning

Reply confidently

Relevant options + pronunciation

Learn afterwards

Saved exchanges → personal practice

Opportunities

The problem: understanding is only half the conversation

A newcomer must also decide what to say, pronounce it and complete a task while the other person waits.

The solution: replies that fit the person and the moment

Combines the current turn, a pre-task plan, activities, preferences and nearby saved-place notes. Learner readbacks update context.

Users and customers across regions

Interstate and international migrants. Customers: migrant associations, NGOs, universities, employers and relocation teams anywhere.

List of features offered by the solution

Hear and respond
Live multilingual speech; three contextual replies.

Say it your way
Express familiar-language intent in the local language.

Recognize learner readbacks
Forgiving repeat matching + session voice estimates.

Plan before leaving
Cab → optional tea → station; Done / Skip / Previous.

Location-linked memory
Opt-in place facts recalled on a fresh return nearby.

Personal memory
Editable activities and preferences; relevant retrieval.

Learn afterwards
Saved exchanges; word bubbles and pronunciation.

Simple, guarded access
One large microphone control; Google sign-in; quotas.

Process flow diagram or Use-case diagram

1. Plan + listen

Optional goal; one large microphone button

→

2. Understand

Live Transcribe produces committed speech

→

3. Route the turn

Readback matching + recent voice clues

Learner repeats a suggested reply

Keep the actual spoken words in context and preserve the options. Continue listening for the other person.

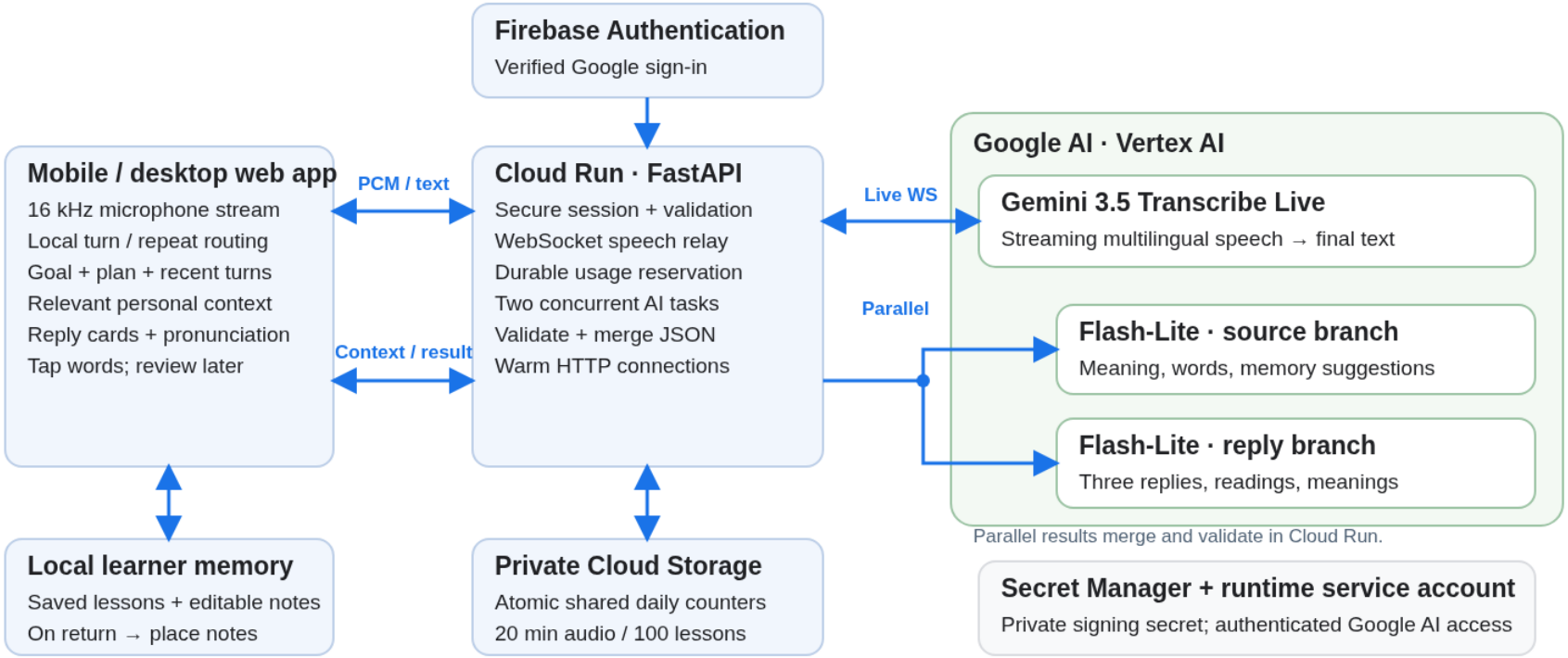
A new turn needs assistance

Parallel Flash-Lite tasks → meaning + three replies → learner speaks → Saved review afterwards.

The learner can correct who spoke. Goals, plan progress and relevant place memories follow the conversation.

Wireframes/Mock diagrams of the proposed solution (optional)

Architecture diagram of the proposed solution



Technologies to be used in the solution

Live speech

Gemini 3.5 Transcribe Live streams multilingual audio while the person speaks.

Fast, economical replies

Two concurrent Gemini 3.1 Flash-Lite tasks: source understanding and reply generation; compact JSON.

Google Cloud

Cloud Run + Vertex AI; Firebase Google sign-in; private Cloud Storage quotas; Secret Manager.

Local personalization

IndexedDB lessons; editable learner profiles, task plans and guarded location-linked memory.

Estimated implementation cost (optional)

~US$5 / learner / month · ₹450

200 listening minutes + 400 new reply turns per month.

100 learners · ~US$465 / month

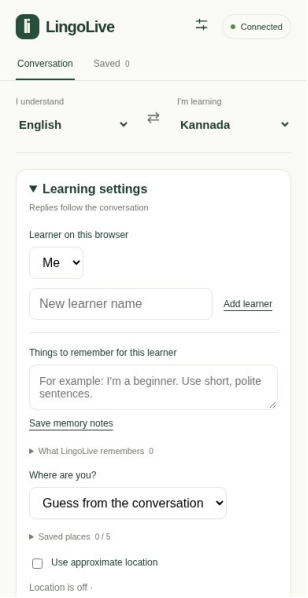
Shared community estimate, including contingency.

Per learner: speech $1.79 + generation $1.76 + shared infrastructure $0.30. Add 20% contingency, then round the budget to US$5.

Usage estimate: 10 minutes and 20 new reply turns per day, on 20 days per month; hosting shared across 100 active learners. Each turn uses two parallel calls.

Rate basis: Live Transcribe $0.008925/min; Flash-Lite $0.25/M input and $1.50/M output. Standard global rates checked 18 Oct 2026; before tax and credits. INR planning rate: ₹90/US$. Detailed assumptions: technical PDF.

Snapshots of the prototype



Plan before leaving



Find a useful reply



Learn afterwards

Recorded prototype UI; Saved view uses sample lesson data. The demonstration includes real Google AI responses.

Prototype Performance report/Benchmarking

Designed for an approximately two-second warm lesson.

1.79 s mean · 10 lessons	Recorded v1.5 warm hosted run; all complete responses succeeded.
2.065 s mean · 4 cases	Later v1.9 Kannada task flows: cab, tea, café and station.
Parallel critical path	Source understanding and replies run together; compact output and warm HTTP.
Local reuse	Readbacks preserve replies; word taps and saved review reuse lesson data.
Memory and access checks	134 backend tests; 16 place-core checks and 8 place browser flows.

Timings measure warm lesson requests; speech also includes transcription finalization and device/network time.

Additional Details/Future Development (if any)

Adoption and business estimates

20 partners × 100 learners = 2,000 learners.

Community-sponsored usage packages.

Estimated operating budget: ~US$5 / active learner / month in a 100-member community.

Partners: migrant groups, NGOs, campuses and employers globally.

Pilot targets and next delivery

Four weeks, 50 learners: ~1,600 encounters and 8,000 assisted turns.

Targets: 70% useful reply sets; 80% task completion.

Next: more language pairs, partner allowances and opt-in local voice enrollment.

Adoption, pilot outcomes and unit economics are planning estimates/targets; assumptions and scale scenario are in the technical PDF.

Provide links to your:

GitHub source — github.com/U4AR/LingoLive

Demo video — 3:17 working demonstration

Final product — live Google Cloud Run prototype

Submission materials — pitch, technical PDF and entry text

Sign in with any verified Google account. A personal Google model key is optional.

Shared daily demo allowance: 20 listening minutes and 100 two-branch lessons; midnight IST reset.

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Thank you

